

# Occupational Musculoskeletal Disorders

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# Prevalence of MS disorder in Taiwan workers

| 身體部位  | 男性              |      |     | 女性              |      |     |
|-------|-----------------|------|-----|-----------------|------|-----|
|       | 患者人數<br>[×1000] | 盛行率  |     | 患者人數<br>[×1000] | 盛行率  |     |
|       |                 | %    | SE  |                 | %    | SE. |
| 脖子    | 372             | 11.9 | 0.4 | 320             | 14.7 | 0.5 |
| 肩膀    | 451             | 14.4 | 0.4 | 379             | 17.4 | 0.5 |
| 上背    | 136             | 4.4  | 0.2 | 117             | 5.4  | 0.3 |
| 下背及腰  | 574             | 18.3 | 0.4 | 428             | 19.7 | 0.5 |
| 手肘    | 183             | 5.6  | 0.3 | 109             | 5.0  | 0.3 |
| 手及手腕  | 335             | 10.7 | 0.4 | 224             | 10.3 | 0.4 |
| 臀及大腿  | 111             | 3.5  | 0.2 | 101             | 4.6  | 0.3 |
| 膝     | 115             | 3.7  | 0.2 | 80              | 3.7  | 0.2 |
| 腳踝    | 100             | 3.2  | 0.2 | 79              | 3.6  | 0.3 |
| 有任一部位 | 1016            | 35.2 | 0.5 | 860             | 39.5 | 0.6 |

# Occupational Back Pain (職業性背痛)

# Back Pain (下背痛)

- The most common reason for filing workers' compensation claims in the United States; accounts for about one fourth of all claims.
- Because the compensation to back pain is generally above the average, about one third of total compensation is paid to back pain claims.
- The second (next to common cold) most common cause of sickness leaves and accounts for about 40% of sick absences from work.

# Back Pain

- The total cost in the United States was estimated to be 50 to 100 billion in 1990.
- Also affects a lot of workers, leads to large numbers of lost workdays, and results in huge economic loss in many countries.
- Cost Taiwan National Health Insurance 3 billion a year.
- 18.9% of Taiwanese worker had back pain in the one-year period before a nationwide survey 1994.
- Only a small number of nation-wide studies have been conducted, and most of them were unable to indicate specific occupational risk factors.

# NHIS-OHS

- In 1988, an Occupational Health Supplement (OHS) was included in the National Health Interview Survey (NHIS) conducted by the US Centers for Disease Control (CDC).
- An analysis estimated that about 22.4 million cases of back pain among workers in the US in 1988 (prevalence=17.6%) lost a total of about 149.1 million work days.
- A large proportion of the cases and the lost workdays were attributable repeated activities at work.

# High Risk Occupation (Men)

| Occupation                               | Cases   | Prevalence |
|------------------------------------------|---------|------------|
| Carpenters                               | 338,000 | 19.2       |
| Automobile Mechanics                     | 173,000 | 17.7       |
| Stock Handlers &<br>Baggers              | 72,000  | 17.2       |
| Plumbers, Pipefitters,<br>& Steamfitters | 91,000  | 16.7       |
| Supervisors, n.e.c.                      | 88,000  | 16.6       |
| Truck Drivers, light                     | 83,000  | 13.0       |

# High Risk Occupation (Women)

| Occupation                                | Cases   | Prevalence |
|-------------------------------------------|---------|------------|
| Nursing Aides, Orderlies,<br>& Attendants | 217,000 | 15.2       |
| Licensed Practical Nurses                 | 85,000  | 13.9       |
| Hairdressers &<br>Cosmetologists          | 91,000  | 12.5       |
| Janitors & Cleaners                       | 94,000  | 12.3       |
| Maids & Housemen                          | 67,000  | 11.9       |
| Laborers,<br>except construction          | 40,000  | 11.1       |

# Sources

- Organic (器官)
- Psychosomatic(官能性):精神因素重要  
(身心症)

# Causes

- Congenital abnormality (先天性畸形).
- Metabolic (代謝性): osteoporosis (骨質疏鬆症) is the most common.
- Degenerative spine diseases (退化性脊椎疾病)
- Trauma (外傷)
- Tumor (腫瘤)
- Inflammatory diseases (發炎性疾病)

# Occupational risk factors

- *Lifting* (搬動及舉起重物)
- *Strenuous work* (費力的工作)
- *Awkward posture* (不自然的姿勢)
- *Whole body vibrations* (全身性的震動)
- *Static posture* (長時間靜態的姿勢)
- *Trauma* (外傷，包括滑倒)

# Non-occupational risk factors

- *Age* (年齡)
- *Female gender* (性別，女性停經後)
- *Obesity* (肥胖), *height*(身高)?
- *Smoking* (抽菸)
- *Exercise* (運動)
- *Psychological factors* (心理因素)

# Lifting

- The most important occupational risk factor.
- NIOSH reviewed 18 major studies and found 13 supporting the association with relative risk generally above 3.
- The association has a dose-response relationship pattern (劑量效應關係).

# Strenuous work

- It puts stress on the spine.
- The effects are smaller than lifting, but most major studies observed the association.

# Awkward posture (不自然姿勢)

- It refers to unnatural posture, including:
  - *Bending* (彎腰；通常是身體往前或往兩旁彎曲) .
  - *Twisting* (扭腰) .
  - *Reaching* (括伸展軀幹) .
  - *Kneeling* (跪姿) .
  - *Squatting* (蹲踞) .

# *Twisting*



# Awkward posture

- Cause back pain directly.
- Also put spine in a vulnerable status and lead to back pain through other causes.
- Bending and twisting were found to have relative risks above 3 in most studies.

# Whole body vibrations (全身性鎮痛)

- Two major types of vibration exposures at work :
  - *Localized (局部性的), such as with hand-hold vibration tools (使用手握電動器械所引起者).*
  - *Whole body (全身性的), affecting the whole body (傳達全身，通常是來自支撐身體的系統，如座椅、工作平台等).*
- The risk is often over-looked, because it is not so obvious.

# Whole body vibrations

- NIOSH reviewed 19 major studies and found 15 supporting the association, but the relative risks had a wide range, depending on the source of vibrations.
- Truck > bus > taxi.

# Static posture

- Prolonged sitting and standing (長時間保持某些靜態的姿勢，也可能與下背痛的發生有關。尤其是坐著及站著而缺乏支撐的姿勢，會對脊椎造成壓力，可能進而引起下背痛) 壓力：躺 < 站 < 坐
- The studies are limited and have no consensus (目前較少有針對這一個危險因子的研究，大多數的資料是對其他因素進行研究時一併收集來的，而且還沒有達成較一致的定論).

# Obesity (肥胖)

- Body weight puts pressure on the spine.
- A review in 1994 found the highest quartile had an increased risk (最肥胖的前四分之一的人口).
- Two large studies (10000s) found associations on specific sub-populations (後來發表的兩個大型上萬個人研究，也僅在特定的族群觀察到兩者的正相關性).

# Smoking

- A review in 1994 on obesity first found the association.
- Another review of literature from 1974 to 1996 (47) found the association was only observed in large studies—small effects.
- Large studies later, especially prospective studies confirm the association.

# Exercise

- Muscle can be trained.
- Exercises may lead to back injuries.
- Even with sport injuries, athletes still have lower risks.

# Psychological factors

- Stress is an independent risk factor (心理因素，尤其是心理壓力，也與下背痛的發生有關，兩者的相關性無法用社會階層或工作的生理負擔解釋，是一項獨立的危險因子).
- Prospective studies confirm the findings (前瞻性研究發現，心理壓力可以用來預測下背痛的發生，而且兩者的相關性很強).

# Symptoms

3  
:  
7 ④ • Acute (急性)

◆ 往往有可以指出的原因。

◆ Chronic (慢性)

◆ 往往是由隱約的不舒適感覺逐漸轉變成為明顯的疼痛；演變的過程可以長達數年，因此常常沒有可以指出的明確原因。

• They may overlap with each other (有些急性發作的個案也會轉變成為慢性的患者，而有些慢性的患者也會有急性發作的時候)

# Symptoms

- May come with other symptoms (除了下背痛本身的症狀之外，有的患者有其他的合併症狀).
- ◆ *HIVD: radiation to legs* (椎間盤脫出經常會有疼痛輻射到兩腿的現象).
- Most occupational back pain cases are chronic.

# Clinical evaluation (評估)

- As in other occupational diseases (與其他職業傷病的診斷一樣).
- ◆ Disorder (病變的存在).
- ◆ Occupational exposure (職業的暴露)
- ◆ Epidemiology evidence (流行病學證據)
- ◆ Temporal relationship (時序性)

# Disorder evidence

- Only a small portion of cases have positive findings in medical image exams (只有少數能夠在醫學檢驗或影像檢查，包括X光、電腦斷層、核磁共振等中找到客觀的病變).
- Mostly depends on history taking and physical examination (主要靠自述症狀及理學檢查，而且是以自述症狀為主).
- Even the most common physical findings such as tender points and limit of range of motions can only be seen in a portion of the patients (常見的理學檢查發現如局部壓痛、活動限制等，也只能在部份患者身上觀察到).

# Occupational exposure

- Risk factors
  - *Lifting: >30kg for above 2h/day or >20kg for above 4h/day.*
  - *Awkward posture (不自然的姿勢).*
  - *Whole body vibrations (全身性的震動).*
  - *Static posture (長時間保持靜態的姿勢) above 4h/day.*
- Temporal relationship: 10 years for HIVD.

# Epidemiology evidence

- Same disorder in co-workers (一起工作的人是否有相同發病情形).
- Recognized as high-risk workers in the literature (文獻上是否有記載病患所從事的工作屬於高危險職業).

# Treatment

- Similar to non-occupational back pain (職業性下背痛的治療與非職業性下背痛的治療大致差不多):
  - ◆ *Rest* (休息).
  - ◆ *Medication* (藥物).
  - ◆ *Physical therapy* (物理治療，冷敷、熱敷、按摩、牽引、束腹等) .
  - ◆ *Surgery* (手術).
  - ◆ *Acupuncture* (針灸).

# Treatment: more important

- Minimize risk factors at work (去除工作中的危險因子).
- Minimize personal risk factors (去除個人的危險因子，包括戒菸、減肥、做適當運動等).

# Prevention

- Avoid or minimize risk factors at work (避免或減少在工作中接觸到危險因子).
- ◆ *Replace manual work with machinery* (運用機械代替或幫助人力從事費力的工作)
- ◆ *Ergonomic design* (使用符合人體工學的工作站、工具、工作流程來減少不自然的姿勢與全身性的振動) ↳ 一天搬2T重以下
- ◆ *Work schedule* (設計適當的作習表、工作量、管理制度來避免長時間保持某些靜態的姿勢).
- ◆ *Rest* (提供足夠的休息、減少工作的心理壓力)

# Prevention

- Health examinations (職前検査、定期健檢)
  - Pre-employment.
  - Periodical.
- Pre-placement training (職前訓練).
- On-job training (在職教育).

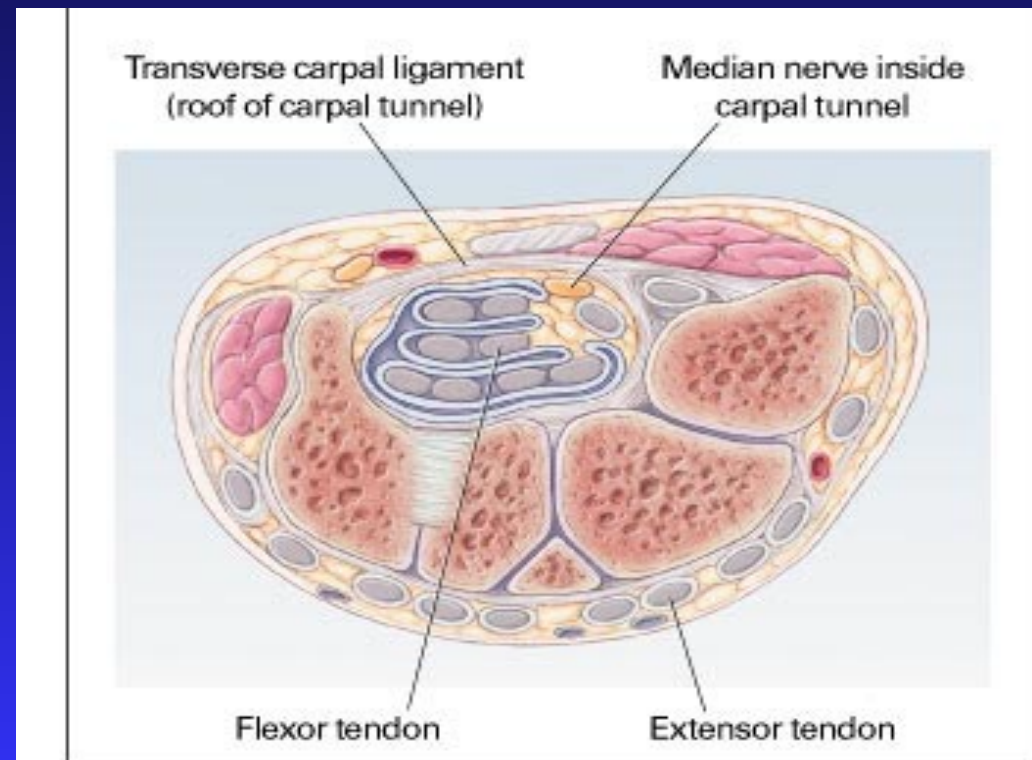
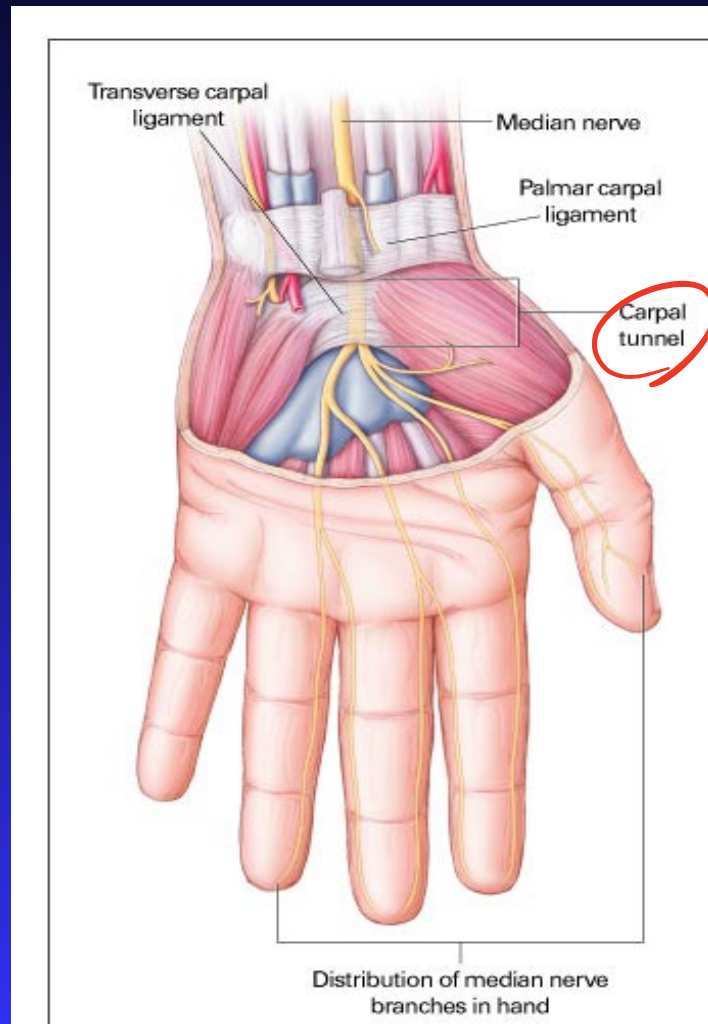
# Occupational Carpal Tunnel Syndrome (職業性腕道症候群)

# 腕道症候群(carpal tunnel syndrome)

- Compression of the Median Nerve (由於正中神經受到壓迫引起).
- Often caused by repetitive motions leading to inflammation of the tendon of flexor muscles (常因手腕的經常反覆性運動，造成前臂屈肌之肌腱發炎所導致)

↳ 肌腱發炎壓迫正中神經

# Carpal tunnel (腕隧道)



# Non-occupational risk factors

- Physical factors

肥胖、懷孕、授乳、口服避孕藥

- Disorders (病)

糖尿病、非特異性關節炎或類風濕性關節炎、甲狀腺機能低下症、自體免疫性疾病、雷諾氏症(Raynaud's disease)

# Non-occupational risk factors

- Trauma (外傷)

直接挫傷正中神經、手腕部骨折癒合不良、骨折或脫臼導致骨片或骨頭突出部壓迫腕道

- Recreational activities (休閒活動)

打電動玩具、彈奏樂器、雕刻

# Occupational risk factors

- Repetitive motion of the wrist (手腕部反覆性動作之作業，尤其是頻率高的反覆性手腕部動作).
- Gripping (用力握緊或抓緊工具或物品之作業).
- Awkward posture of hand or wrist (手部或腕部以不自然姿勢操作之作業，如彎曲、過度伸展，或向尺骨、撓骨側彎曲等的動作).
- Direct compression (直接對腕迅施予壓迫).
- Hand-hold vibration tool (振動性手工具).

# Taiwan Diagnostic Criteria

## (我國職業性腕道症候群診斷基準)

- Symptoms (於正中神經所支配的手掌區域出現感覺功能異常或疼痛麻痺之自覺症狀), Signs (理學檢查出現Tinel's sign或Phalen's sign陽性), and delay nerve conducting velocity(神經電氣學檢查確實證明腕道症狀群存在病變的存在)
- Exclusion of non-occupational causes (能排除各種常見非職業性致病因素或腕部傷害職業的暴露).
- Temporal relations (發病時間在開始具危害性作業或工作之後才發生時序性).
- Risk factors at work (工作有職業性相關暴露)

# High risk occupations

- . 打字員
- . 鍵盤操作員
- . 生活用品零售之算帳員
- . 包裝工人
- . 焊接作業
- . 切削工
- . 打蠟工
- . 電線電纜工
- . 排字工人
- . 鑿石工人
- . 礦工
- . 牙醫師
- . 服裝製造業員工
- . 音樂演奏者
- . 汽車裝配工
- . 雜貨整理業之勞工

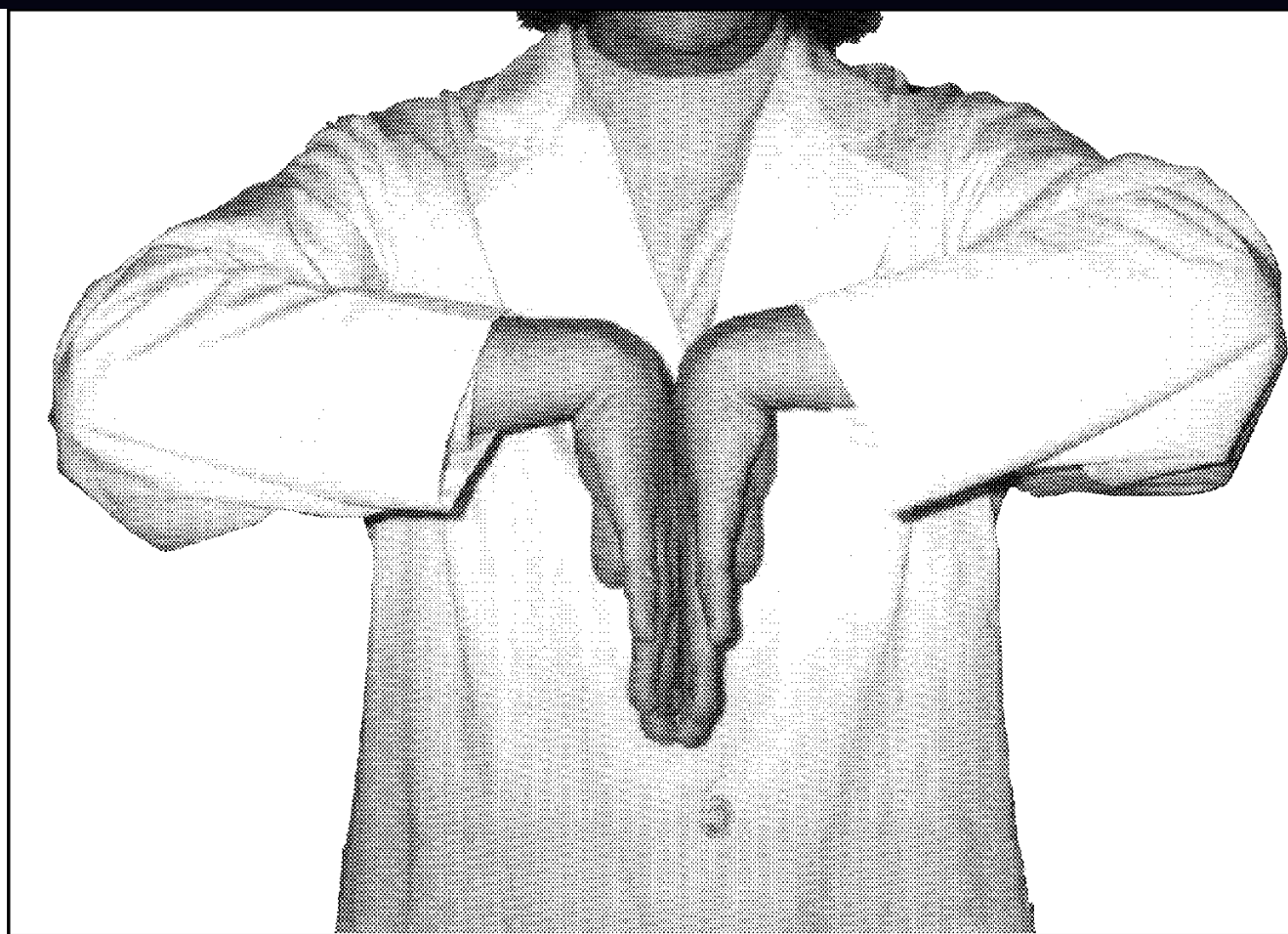
- . 屠宰業
- . 冷凍食品作業工人
- . 木工
- . 伐木工
- . 油漆工
- . 裝潢工人
- . 鋸木工
- . 水泥工
- . 洗衣房工作者
- . 搬運工人
- . 麵包師傅
- . 管家或廚師
- . 縫繡工業員工
- . 電子零件裝配工
- . 機具裝配工
- . 寶石切割、雕琢工人

# Clinical presentation (臨床表徵)

- Abnormalities of the first to fourth digits (手掌部位正中神經支配之區域(大拇指、食指、中指及近大拇指側的第四指一半等區域)，出現感覺功能的異常，通常以中指與食指之遠端區域(如指尖)較常發生異常。
- Include hyperesthesia (感覺過敏), paresthesia (感覺異常，如灼熱感、針刺感等), numbness (感覺遲鈍), and pain (疼痛).
- Often observed in the dominant hand (通常會先出現於慣用手) and wake up the patient at night (在晚上特別容易發生症狀，有時會因疼痛或麻痺而自睡眠中醒來).

# Examination

- Tinel's sign (Tinel's test)
  - Sensitivity (敏感度): 38% to 100% (?).
  - Specificity (特異性): 55% to 100% (?).
- Phalen's sign (Phalen's maneuver; Phalen's test)
  - Sensitivity (敏感度): 42% to 92% (?).
  - Specificity (特異性): 33% to 98% (?).



F.S.

圖 4-2 法蘭氏（Phalen）測試

30-60 sec.

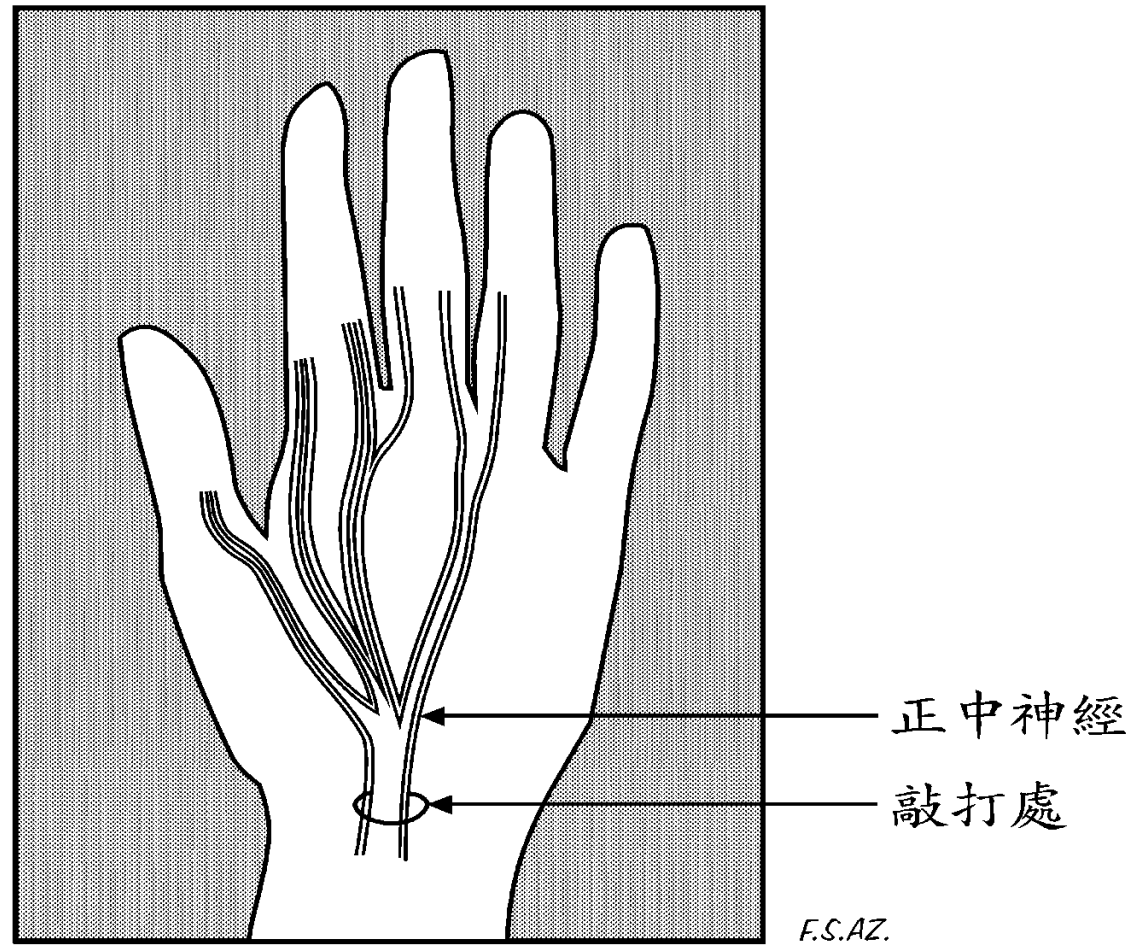


圖 4-1 提內爾氏 (Tinel) 徵象

# Nerve conducting velocity (神經電生理學檢查)

- The golden standard of the diagnosis.
- The normal range varies across laboratories (特別注意的足測定時的皮膚溫度會影響檢查結果，因此最好每個實驗室建立自己的標準測定步驟，控制受測者皮膚表面溫度於一定範圍，並且上建立一套自己實驗室分年齡的正常值範圍，以利結果之判定)
- May identify malingering (裝病).

# NCV

- May accompanied with EMG (必要時亦可進行肌電圖檢查).
- Sensitivity (敏感度) 50~85%, specificity (特異性) > 95%.

# Treatment

- Fixation (固定腕部:對於腕部必須反覆前彎或背彎之勞工，將腕部保持直的狀態，十分重要，可以避免對腕道造成更大的壓力，避免病情的惡化。因此以腕部固定片固定腕部為重要之治療方針。若未能固定音，則必須避免會增加腕道壓力的動作或工作。)
- Medication (藥物治療:口服或局部使用非類固醇抗發炎劑，可減少疼痛及炎症之反應。)

# Treatment

- Local injection (使用皮質類固醇作腕道內局部注射，可得到較快的症狀緩解，但必須小心避免副作用之產生).
- Surgery (手術治療:若症狀嚴重。如併有肌肉萎縮，在以上治療均未能得到適宜之緩解時，可以手術將腕道包膜打開，解除正中神經之壓力).

# Tennis Elbow

(職業性肱骨外側上踝炎，網球肘)

# Causes

- Wrist extensors are repetitively contracted.
- Chronic overload of the bone-tendon junction, leads to changes at this junction.
- Microruptures occur, especially in *extensor carpi radialis brevis* (ECRB; 橈側伸腕短肌 ).

# Prevalence of lateral epicondylitis

- 5.8% in a rural population.
- 7.5% in an urban population.
- 14% in workers with jobs demanding repetitive resisted elbow and wrist extension.
- In the occupational setting, 11.7 in 10,000 yearly.

Darmawan J, Valkenburg HA, Muirden KD, et al. The prevalence of soft tissue rheumatism. A WHO-ILAR COPCORD study. *Rheumatol Int.*

1995;15:121–124.

Ritz BR. Humeral epicondylitis among gas- and waterworks employees. *Scand J Work Environ Health.*

1995;21:478–486.

# Four stages of tendinosis

- Stage I: transient inflammatory changes, which often resolve.
- Stage II: temporary pathologic changes associated with tendon degeneration, most common.
- Stage III: permanent changes involved in tendinosis, with complete rupture of the tendon.
- Stage IV: has characteristics of stages II or III, with associated calcification or fibrosis.

Nirschl RP: Tendinosis of the elbow (tennis elbow). J Bone Joint Surg 1999, 81:259–278.

# Diagnosis

- Epicondylar pain and tenderness, and pain on resisted extension of the wrist.
- Good interobserver consistency both in hospital and epidemiologic settings.

Palmer K et al. The Southampton examination schedule for the diagnosis of musculoskeletal disorders of the upper limb. *Ann Rheum Dis.* 2000;59:5–11.

Walker-Bone K et al. Reliability of the Southampton examination schedule for the diagnosis of upper limb disorders in the general population. *Ann Rheum Dis.* 2002;61:1103–1106.

# Risk factors for tennis elbow

- Carry out activities such as gripping and twisting.
- In fixed or awkward position.
- With constant repetition.
- With excessive force.
- Without enough recovery from the wear and tear.
- Other possible risk factors.
  - Age
  - Female gender

# Treatment options of tennis elbow

- Rest from the activities that cause elbow pain.
- Medication to reduce the inflammation.
- Physical therapy.
- Local steroid injection.
- Botulinum toxin injection.
- Surgical treatment if conservative treatment fails.

Sevier TL, Wilson JK. Treating lateral epicondylitis. Sports Med. 1999; 28:375–380.

# Treatment of tennis elbow

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*Injections*

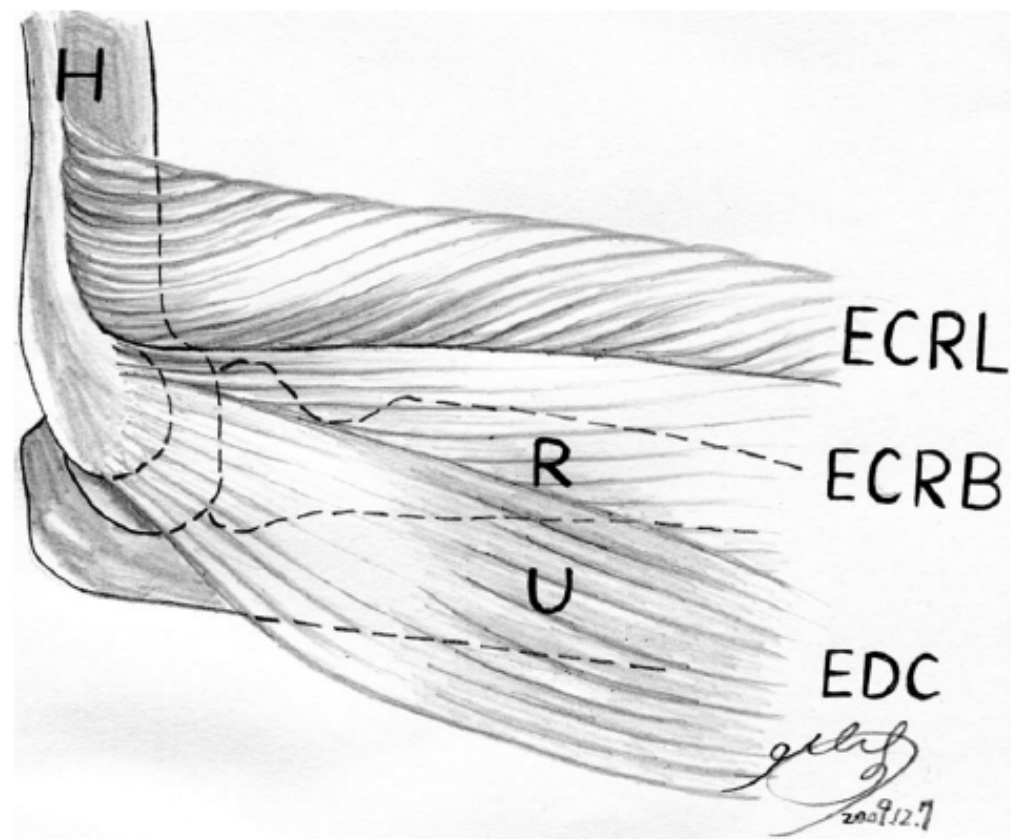
## ORIGINAL RESEARCH ARTICLE

# Comparison Between Botulinum Toxin and Corticosteroid Injection in the Treatment of Acute and Subacute Tennis Elbow

A Prospective, Randomized, Double-Blind, Active Drug-Controlled Pilot Study

## ABSTRACT

Lin Y-C, Tu Y-K, Chen S-C, Chen S-S, Lin I-L, Guo H-R: Comparison between botulinum toxin and corticosteroid injection in the treatment of acute and subacute tennis elbow: A prospective, randomized, double-blind, active drug-controlled pilot study. *Am J Phys Med Rehabil* 2010;89:653–659.



**FIGURE 1.** *The injection was administered into the ECRB muscle near the common origin of the wrist and finger extensors at the lateral epicondyle. H, humerus; R, radius; U, ulna; ECRL, extensor carpi radialis longus; ECRB, extensor carpi radialis brevis; EDC, extensor digitorum communis.*

# Treatment of tennis elbow

**TABLE 1** Baseline characteristics of participants in the Botox and steroid groups

| Characteristic                | Botox Group<br>( <i>n</i> = 8) | Steroid Group<br>( <i>n</i> = 9) | <i>P</i> <sup>a</sup> |
|-------------------------------|--------------------------------|----------------------------------|-----------------------|
| Gender, <i>n</i>              |                                |                                  |                       |
| Men                           | 3                              | 6                                | 0.35                  |
| Women                         | 5                              | 3                                |                       |
| Affected side, <i>n</i>       |                                |                                  |                       |
| Left                          | 5                              | 4                                | 0.64                  |
| Right                         | 3                              | 5                                |                       |
| Symptoms, <i>n</i>            |                                |                                  |                       |
| >3 mos                        | 0                              | 1                                | >0.99                 |
| <3 mos                        | 8                              | 8                                |                       |
| Occupation, <i>n</i>          |                                |                                  |                       |
| Nonforceful manual work       | 5                              | 4                                | 0.64                  |
| Forceful manual work          | 3                              | 5                                |                       |
| Age (yrs), mean (SD)          | 45.9 (7.8)                     | 44.6 (11.0)                      | 0.53                  |
| VAS (mm), mean (SD)           | 45.0 (27.3)                    | 57.4 (18.7)                      | 0.27                  |
| Grip strength (kg), mean (SD) | 27.5 (9.4)                     | 35.2 (13.8)                      | 0.21                  |
| WB-D1, mean (SD)              | 13.6 (2.0)                     | 12.9 (1.7)                       | 0.37                  |
| WB-D2, mean (SD)              | 13.1 (1.9)                     | 12.6 (1.6)                       | 0.48                  |
| WB-D3, mean (SD)              | 14.0 (1.4)                     | 14.3 (1.5)                       | 0.62                  |
| WB-D4, mean (SD)              | 13.1 (1.8)                     | 13.2 (2.5)                       | 0.92                  |

Data are number or mean (SD).

<sup>a</sup>*P* for Fisher's exact test or Wilcoxon's rank-sum test.

WB-D1, WB-D2, WB-D3, WB-D4: World Health Organization Quality of Life Brief Questionnaire domain 1, domain 2, domain 3, and domain 4; VAS, Visual Analogue Scale.

# 職業性肌肉骨骼傷害的個人預防

## (General rules of preventing MS disorders)

- 量力而為。(Don't over-work.)
- 使用工具協助工作。(Use tool.)
- 避免不自然的姿式。(Avoid awkward postures.)
- 避免長時間的固定姿態。(Avoid static postures.)
- 適當的休息。(Proper resting.)
- 適當的運動與鍛練。(Proper exercise.)